

Bank of Leeds Memorandum

Yorkshire Water Export Initiative

A Strategic Proposal for Bulk Freshwater Exports to the Gulf Cooperation Council (GCC)

Prepared for: Bank of Leeds Investment Committee

Executive Summary

This memorandum proposes the creation of a **Yorkshire Water Export Company (YWEC)** to develop a bulk freshwater export industry serving the Middle East.

The concept is simple:

Oil tankers arrive in Europe carrying crude oil.

They frequently return with little or no commercial cargo.

Instead of returning empty, suitably prepared vessels (or dedicated water tankers operating on backhaul routes) would transport treated Yorkshire freshwater to Gulf nations where water scarcity is a strategic issue.

Rather than exporting a manufactured product, Yorkshire would export one of Britain's most abundant natural resources:

High-quality freshwater.

Strategic Opportunity

Yorkshire receives substantial annual rainfall and possesses extensive upland reservoir systems.

Meanwhile, Gulf countries depend heavily on:

- desalination
- groundwater
- imported food
- water-intensive energy infrastructure

Freshwater demand continues to increase because of:

- population growth
- industrial expansion
- agriculture
- climate pressures

Although desalination remains the dominant supply source, imported water can make economic sense for premium industrial or municipal uses if delivered efficiently on return shipping routes. Historical studies estimated backhaul transport costs of around US\$1–2 per cubic metre under favourable conditions, comparing competitively with some desalination costs of the period.

Proposed Export System

Stage One

Water is sourced from:

- Yorkshire reservoirs (subject to environmental licensing)
- surplus winter abstraction
- treated potable supplies dedicated for export

Water is transported via pipeline to:

Humber Export Terminal

A new loading facility located on the Humber Estuary.

Features:

- storage reservoirs
 - filtration
 - UV sterilisation
 - tanker loading systems
 - customs facilities
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Shipping Model

Rather than commissioning an entirely new fleet, the proposal assumes contracts with tanker operators for return voyages where technically and legally suitable.

Typical large tanker capacity:

Approximately **250,000 cubic metres**

Equivalent to:

250 million litres

Export Programme

Assume:

40 export voyages annually

Annual exports:

10 million cubic metres

Equivalent:

10 billion litres

Financial Model

Water Sales Price

Illustrative wholesale export price:

£2.00 per cubic metre

Annual export volume:

10 million cubic metres

Annual revenue:

£20 million

Operating Costs

Item	Annual Cost
Water treatment	£2m
Pipeline operation	£2m
Port operations	£3m
Shipping contribution	£5m
Administration	£2m
Total:	

£14 million

Operating Profit

Revenue:

£20 million

Operating costs:

£14 million

Estimated operating surplus:

£6 million annually

Initial Capital Investment

Project	Cost
Humber loading terminal	£25m
Storage reservoirs	£12m
Treatment plant	£15m
Pipelines	£20m
Environmental monitoring	£5m
Working capital	£8m

Total Investment

£85 million

Economic Benefits

Employment

Construction:

Approximately **500 jobs**

Operations:

Approximately **120 permanent jobs**

Additional employment:

- engineering
 - logistics
 - marine operations
 - environmental management
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Yorkshire Supply Chain

The project would support:

- pipeline contractors
 - civil engineering firms
 - port operators
 - shipping services
 - laboratory testing
 - environmental monitoring
 - digital infrastructure
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Environmental Considerations

Any commercial export scheme would require careful regulation to ensure that exports only use **surplus water** after protecting:

- public drinking water supplies
- river flows
- reservoir resilience
- biodiversity
- future drought security

The project would therefore operate only when hydrological conditions permit, with export volumes varying by season.

Risks

Key risks include:

- environmental permitting
- long-term water availability
- customer demand
- tanker availability
- water quality compliance
- international regulations governing carriage of potable water

The concept would require detailed engineering and legal work before implementation.

Strategic Value

For the Bank of Leeds, the project would diversify Yorkshire's export economy into an unconventional but potentially valuable niche.

It would:

- create export revenues
 - strengthen the Humber ports
 - generate engineering and logistics employment
 - establish Yorkshire as a supplier of high-value environmental resources
 - build expertise that could later support exports of water treatment technology and environmental engineering services
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Recommendation

It is recommended that the Bank of Leeds commission a **£2 million feasibility study** to assess:

- sustainable water availability
- engineering requirements
- port infrastructure
- tanker conversion and charter options
- environmental impacts
- Gulf customer demand
- long-term commercial viability

The study should determine whether Yorkshire could develop a commercially and environmentally sustainable freshwater export industry using available maritime backhaul capacity while safeguarding domestic water resources.